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Comparing the analgesic utility & safety of erector spinae plane block versus thoracic epidural for multiple rib fracture trauma: a retrospective cohort analysis

Nishant Merchant [a], Douglas J. MacLean [b], Sarah R. Vincze [c,*,1], Jyoti Chhabra [d], Aseel Walker [c], Kevin Finkel [c]

[a] Department of Acute Care Surgery, Hartford Healthcare Medical Group, Hartford, USA
[b] Frank H. Netter School of Medicine at Quinnipiac University, North Haven, USA
[c] Department of Anesthesiology, Integrated Anesthesia Associates, Hartford, USA
[d] Department of Research, Hartford Hospital, Hartford, USA

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ABSTRACT

Purpose: Despite long-standing use of thoracic epidural block (TEB) for multiple rib fracture trauma, emerging research suggests that the erector spinae plane block (ESPB) offers similar analgesia and increased safety benefits. This study compared the analgesic effectiveness and safety of TEB versus ESPB for multiple rib fracture trauma.

Methods: A historical cohort analysis was conducted, including 324 adult patients who received continuous TEB (n = 188) or ESPB (n = 136) plus multimodal analgesia following multiple rib fracture trauma. The primary objective compared average numeric pain scale (NPS) scores at rest and with activity over the first five days of local anesthetic infusion. Other outcomes included opioid consumption, length of stay data, venous thromboembolism (VTE) prophylaxis time measures, as well as vasoactive support, complication, and mortality rates.

Results: Average pre- and post-block NPS scores at rest were greater in the TEB group [5.6 (2.1) vs. 4.8 (2.2), $p = 0.002$; 48–72 h: 3.7 (2.2) vs. 2.9 (2.2), $p = 0.002$, respectively). However, the daily changes in NPS scores throughout anesthetic infusion were similar between groups, without significant difference in opioid consumption. First VTE prophylaxis was given prior to block placement more frequently in the ESPB group (66.9 vs. 31.4%, $p < 0.001$), and median time between pre and post block VTE prophylactic doses was shorter [12.6 (13.7) vs. 23.9 (12.3) hours, $p < 0.001$].

Conclusion: Continuous TEB and ESPB demonstrated similar analgesic efficacy for multiple rib fracture trauma. The ESPB group received earlier VTE prophylaxis without disruption in dose administration for block placement, supporting enhanced safety of ESPB.

Introduction

Traumatic rib fracture injuries account for an estimated 48,000 annual hospital admissions in the US alone [1], and can cause debilitating pain, contributing to enhanced morbidity and mortality in affected patients [2]. Inadequate pain management increases the risk of respiratory complications and development of chronic pain [3], emphasizing the importance of early and optimal implementation of multimodal analgesia. Common analgesic interventions for rib fracture trauma include continuous infusions of local anesthetics through either thoracic epidural block (TEB) or plane blocks paired with multimodal analgesics [3,4]. TEB is a long-established standard of neuraxial analgesia for traumatic rib injuries that effectively blocks motor and sensory nerve roots in the epidural space as they leave the spinal cord to prevent and minimize pain. Compared to intravenous opioids, TEB has been associated with reduced pain scores, and shorter intensive care unit (ICU) stay among patients with traumatic rib fractures [5]. Even so, recent literature has increasingly criticized TEB for its technical complexity, association with adverse effects such as hypotension, urinary retention, and hematoma, and its clinical contraindications such as

* Correspondence to: Department of Anesthesiology, Integrated Anesthesia Associates, 100 Retreat Avenue, Suite 900, Hartford, CT 06106, USA.
E-mail address: sarah.vincze@hhchealth.org (S.R. Vincze).
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patients on therapeutic or prophylactic anticoagulants, coagulopathy, or spinal fracture which commonly occur with rib fracture trauma [6,7], a significant portion of the rib fracture trauma patient population require alternative pain management.

Continuous regional anesthetic blocks such as the erector spinae plane block technique (ESPB) have grown favor among providers in recent years as safe and effective alternatives for rib fracture pain management. ESPB is an inter-fascial plane block involving injection and continuous infusion of local anesthetic between the erector spinae muscle and the transverse processes of the thoracic vertebrae targeting the dorsal rami of the spinal nerves thus providing analgesia to the thoracic wall [8]. In contrast to TEB, the site of ESPB is further away from essential nerves, blood vessels, and internal structures, reducing the risk of damage and bleeding complications in these areas. Additionally, ESPB and catheter placement are usually performed under ultrasound guidance, enhancing the safety profile compared to TEB. Numerous clinical studies have associated the use of continuous plane blocks for rib fracture trauma with significantly reduced pain and opioid consumption as well as improved respiratory function with minimal complications and without interference or interruption of anticoagulation therapy [9–15].

Existing prospective studies comparing the efficacy and safety of regional nerve/plane blocks versus TEB have reported similar pain scores, analgesic requirements, and respiratory function among trauma patients [14,16–21]. However, several of these randomized controlled trials (RCTs) reported heightened incidence of hypotension and hemodynamic instability among patients who received TEB versus ESPB [16,20,21]. Due to the randomized nature of these trials, patients were commonly excluded from these studies for spinal fracture or preexisting spinal issues, intubation or ventilation requirement, hemodynamic instability, pre-existing coagulopathy or anticoagulation use, and/or older age (>60 to >75) depending on the trial). Though important to maximize the safety of trial participants, these exclusion criteria limit the generalizability of these findings to the overall rib fracture trauma population. This retrospective cohort analysis, therefore, sought to compare the clinical effectiveness of ESPB to TEB for analgesic management of rib fracture trauma in a broad patient population, accounting for these important clinical factors. We hypothesized that implementation of ESPB for rib fracture trauma, when compared to TEB, would provide non-inferior analgesia with reductions in opioid requirements, hospital length of stay, and complication incidence.

Methods

Design

This study was approved by local Institutional Review Board (February 2022) and carried out according to institutional and Good Clinical Practice guidelines. Given minimal risk to study participants, a total waiver of written informed consent was granted. This report has been written in accordance with the Strengthening Reporting of Observational Studies in Epidemiology guidelines [22]. This single-center, historical cohort analysis included adult patients with multiple rib fracture trauma admitted for care between January 2017 and December 2021. Patients were identified for study inclusion via the institution's thoracic trauma registry. All additional data were collected retrospectively from eligible patients' electronic medical records (EMR).

Population

Adult patients with no upper age limit admitted to our level-one trauma center (USA, American College of Surgeon's designation) with multiple, unilateral rib fracture trauma, with an Abbreviated Injury Score (AIS) ≥ 2 for thoracic trauma and < 2 for associated non-thoracic trauma who received multimodal analgesia (MMA) were included. MMA was defined as the use of at least two non-opioid analgesic drugs administered by the same or different routes including non-steroidal anti-inflammatory drugs, muscle relaxants, gabapentin, and local anesthetics. Patients were divided into TEB or ESPB grouping, depending on the block technique administered.

Patients that presented to the emergency department (ED) without hospital admission, admission for less than half a day, or expiration within 48 h of admission were excluded from analysis. Patients that received a different regional anesthesia technique than continuous TEB or ESPB, or both techniques within the same admission were excluded to eliminate confounding analgesic influence. Parasternal blocks are frequently used to manage pain from sternal fractures, thus any patients with additional sternal fracture were excluded. Patients with one or multiple non-thoracic traumas (AIS>2) and those who underwent combined surgery in both thoracic and abdominal cavities were excluded due to their confounding influence on pain and recovery outcomes assessed. Likewise, patients presenting with bilateral rib fractures were excluded from analysis due to the expected greater level of pain compared to unilateral fracture, in addition to its association with increased risk of severe, internal organ trauma [23]. Patients with multiple thoracic trauma admissions within the study period were included for the first occurrence only.

Data Collection

Demographics & baseline characteristics

Demographic variables included age, sex, race, ethnicity, and body mass index (BMI). Baseline health characteristics at admission included chronic comorbidity burden, severity of trauma, and rate of surgical intervention. To compare the extent and severity of chronic pre-existing disease, Charlson Comorbidity Index (CCI) scores were reported [24], while injury Severity Score (ISS) and rate of surgical intervention were used to quantify total acute trauma severity [25]. AIS and frequency of flail versus non-flail rib fracture diagnosis depicted severity of rib fracture trauma [26].

Outcomes

The primary outcome comprised mean numeric pain scale (NPS) scores at rest and with activity during the first five days of anesthetic infusion. Total opioid consumption was also quantified in the first five days of infusion and presented as median morphine milligram equivalents (MME); all means of opioid use were accounted for in the calculation of opioid consumption including patient-controlled analgesia (PCA) and continuous infusions. Length of stay (LOS) data including total time in the ED, ICU, and hospital duration were reported as median hours or days. Incidence of hemodynamic instability was identified based on requirement of vasoactive support. Instability occurring within 24 h of TEB placement without pre-existing hypotension or attribution to procedural intervention was considered TEB-related. Any vasoactive support within the ESPB group was attributed solely to trauma, as hemodynamic instability is not a recognized complication of ESPB. Time to first venous thromboembolism (VTE) prophylaxis and time between pre and post block placement VTE prophylaxis doses were compared between groups given the necessity to pause anticoagulation for epidural placement. Other outcomes included time to first ambulation and time to regional anesthesia placement from admission, as well as rates of ventilator and tracheostomy intervention. Frequency of block-related complications and other hospital- or trauma-related complications were reported. Block-related complications included bleeding or hematoma, local anesthetic systemic toxicity (LAST), and neuropathy. Hospital or trauma-related complications included acute respiratory distress syndrome (ARDS), delirium, encephalopathy, infection, pneumonia, and VTE. Finally, rates of mortality (in-hospital and within 30 days of discharge) were compared.

Statistical analyses

Stem-and-leaf plots were used to determine normality of continuous data. Student's T tests were used to compare continuous, normally distributed data, depicted by mean and standard deviation (SD). Non-parametric continuous data and ordinal variables were presented using median and interquartile range (IQR) and compared with Mann-Whitney U tests. Categorical data were described using counts and frequencies and compared with Pearson's Chi-Square tests. An alpha level less than 0.05 denoted statistical significance. Multivariate regression analyses were performed for any statistically significant findings to identify significant predictive factors; in addition to study group, baseline demographic or health characteristics were included in the regression analyses based on clinical relevance and presence of statistically significant differences between groups. Linear regression was used for assessment of continuous, normally-distributed data, whereas, logistic regression was applied for continuous, nonparametric data. Cox regression was used for 'time to' outcomes. To check for co-linearity between predictor variables, correlation matrixes were generated; a conservative Pearson correlation coefficient greater than 0.8 was used as a threshold for co-linearity. All statistical testing was carried out using SPSS version 29 (IBM Corp. Released 2023. IBM SPSS Statistics for Windows, Armonk, NY). This data analysis and statistical plan was written and filed with the local institutional review board before data were accessed.

Results

A total of 324 adult patients admitted with rib fracture trauma between 2017 and 2021 were included for analysis, stratified by regional anesthesia technique: TEB (n = 188) or ESPB (n = 136) (Fig. 1). On average, the ESPB group was older than the TEB group [70 (17) vs. 65 (17) years, $p < 0.01$] and exhibited a higher rate of pre-admission anticoagulant use (28.7 vs. 3.7%, $p < 0.001$) (Table 1). There were no differences in mean AIS and ISS scores, rates of flail/non-flail injury, or surgical intervention during admission. Median block duration was similar between groups [4.6 (2.9) vs. 4.7 (2.7), $p = 0.91$].

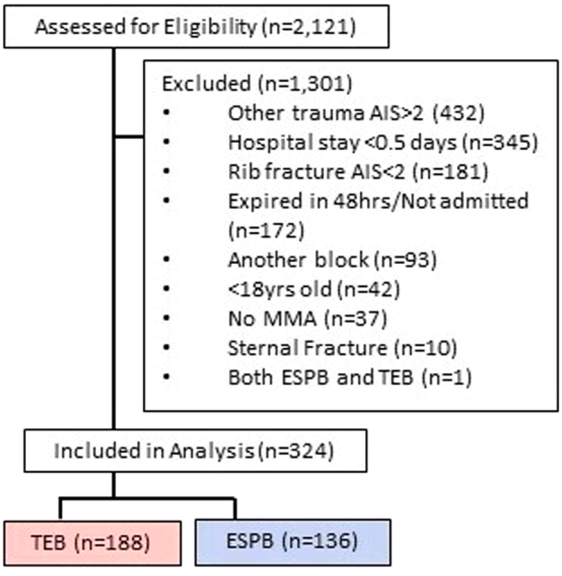


Fig. 1. CONSORT Diagram, Abbreviations: AIS, Abbreviated Injury Scale; MMA, Multi-Modal Analgesia; ESPB, Erector Spinae Plane Block; TEB, Thoracic Epidural Block.

Table 1
Baseline Patient Characteristics.

Characteristic	TEB (n = 188)	ESPB (n = 136)	p-value[A]
Age, years, mean (SD)	65 (17)	70 (17)	**< 0.01**
Patients ≥ 65 years old, n (%)	96 (51.1)	89 (65.4)	**0.01**
Sex, n (%)	72 (38.3)	54 (39.7)	0.89
Female	116 (61.7)	82 (60.3)	
Male			
Race, n (%)	5 (2.7)	4 (2.9)	0.63
Black or African American	26 (13.8)	14 (10.3)	
Other[B]	157 (83.5)	118 (86.8)	
White			
Ethnicity, n (%)	23 (12.2)	9 (6.6)	0.11
Hispanic or Latino	164 (87.2)	124 (91.2)	
Not Hispanic or Latino	1 (0.5)	3 (2.2)	
Unknown			
BMI, mean (SD)	27.9 (6.3)	27.9 (6.1)	0.98
CCI, median [IQR]	0 [1]	0 [1]	0.29
AIS Rib Fractures, mean (SD)	2.8 (0.4)	2.9 (0.4)	0.09
Rib Fracture Injury, n (%)	25 (13.3)	12 (8.8)	0.28
Flail	163 (86.7)	124 (91.2)	
Non-Flail			
Lumbar Spinal Fracture[C]	17 (9.0)	14 (10.3)	0.71
ISS, mean (SD)	16.5 (7.4)	15.7 (6.8)	0.33
Pre-Admit Anticoagulation, n (%)	7 (3.7)	39 (28.7)	**< 0.001**
Pre-Block Hemodynamic Instability, n (%)	16 (8.5)	19 (14.0)	0.12
Surgical Intervention, n (%)	61 (32.4)	44 (32.4)	0.92
Block Duration, days, median (IQR)	4.6 (2.9)	4.7 (2.7)	0.91
≥ 5 days, n (%)	78 (41.5)	56 (41.2)	
< 5 days, n (%)	110 (58.5)	80 (58.8)	

[A]Bolded p-value represents significant finding at the 0.05 level. [B]Other Race includes Asian, American Indian, Hawaiian or Other Pacific Islander, and Unknown categories. [C]Transverse process or minor vertebral body without cord involvement. Abbreviations: TEB, Thoracic Epidural Block; ESPB, Erector Spinae Plane Block; SD, Standard Deviation; Body Mass Index; CCI, Charlson Comorbidity Index; IQR, Interquartile Range; AIS, Abbreviated Injury Scale; ISS, Injury Severity Score.

Primary outcome

Mean pre-block NPS score at rest was greater among the TEB group [5.6 (2.1) vs. 4.8 (2.2), $p < 0.01$] and remained greater across the first five days of continuous anesthetic infusion, specifically between 48 and 96 h [3.7 (2.2) vs. 2.9 (2.2), $p < 0.01$; 3.5 (2.2) vs. 2.6 (2.2), $p < 0.01$, respectively] (Fig. 2A). Mean pain with activity was also greater in the TEB group, reaching significance between 48 and 96 h [4.5 (2.4) vs. 3.9 (2.4), $p = 0.04$; 4.2 (2.5) vs. 3.5 (2.5), $p = 0.02$, respectively] (Fig. 2B). However, the mean deltas in pain at rest and with activity over the five-day infusion period were similar among groups (Supplementary Table 1).

Secondary outcomes

Five-day MME-adjusted median opioid consumption was similar between study groups in the first five days of continuous anesthetic infusion (Fig. 3). PCA opioid infusions were more common among the TEB group, though the difference was not statistically significant (21.8 vs. 15.4%, $p = 0.185$). All other secondary outcome data are described in Table 2 Median ED LOS was longer in the ESPB group, though there were no significant differences in ICU or hospital LOS. Rate of vasoactive support during admission was similar between groups; however, 13 patients (6.9%) in the TEB group required vasoactive support due to block-induced hypotension. There were no statistically significant differences in time to first ambulation after block placement, ventilator or tracheostomy requirement, complications or mortality rates. Time to first VTE prophylaxis was similar between groups, though first VTE prophylaxis before block placement was more common in the ESPB group (66.9 vs. 31.4%, $p < 0.001$). Median time to regional anesthesia

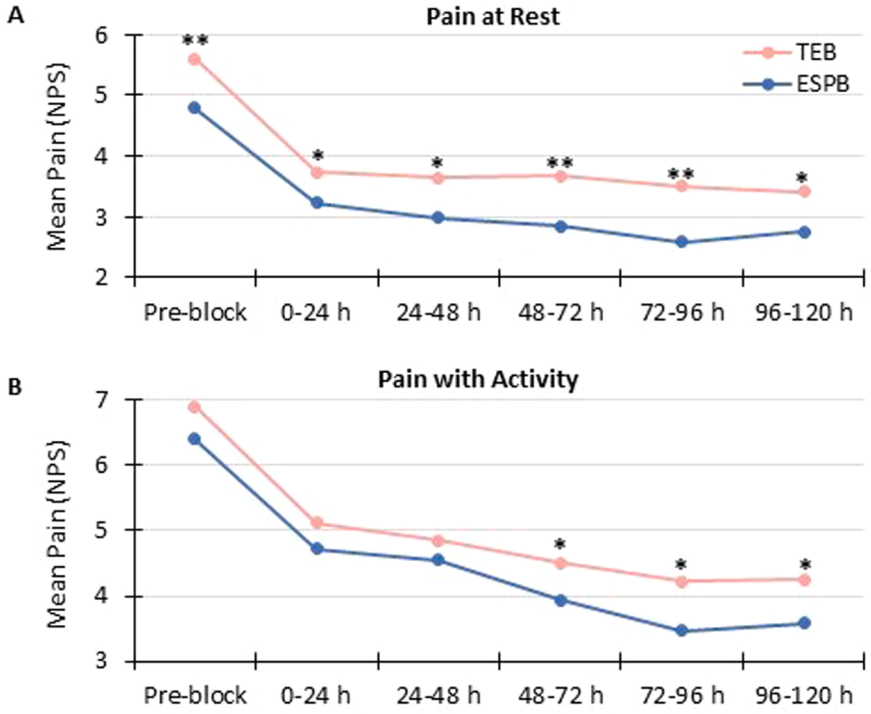


Fig. 2. Primary outcome: mean NPS score from admission (pre-block) through the first five days of continuous anesthetic infusion A) at rest B) with activity, *p < 0.05, **p < 0.01, p < 0.001***. Abbreviations: TEB, Thoracic Epidural Block; ESPB, Erector Spinae Plane Block, NPS, Numeric Pain Scale.

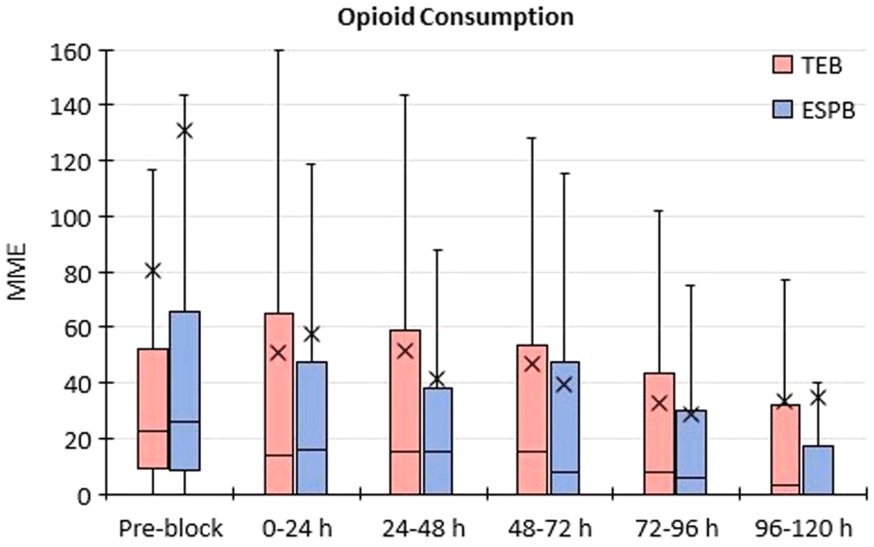


Fig. 3. MME opioid consumption from admission (pre-block) through the first five days of continuous anesthetic infusion. Within each box and whisker plot, the midline represents median while 'x' depicts the mean. Abbreviations: MME, Morphine Milli-Equivalents; h, hours.

block placement was shorter in the TEB group [17.2 (18.3) vs. 25.0 (29.1) hours, p < 0.001]. Median time between VTE prophylactic doses for regional anesthesia placement was shorter in the ESPB group [12.6 (13.7) vs. 23.9 (12.3) hours, p < 0.001]. Though not statistically different, four (2.1%) VTE events were identified in the TEB group versus none in the ESPB group.

Multivariate regression analysis

Mean patient age was significantly different between TEB and ESPB groups, and thus included as a covariate in all regression models to elucidate its confounding influence, if any, on study outcomes. For analgesic outcomes including NPS pain and total MME opioid use, pre-block mean pain was included as a covariate to account for significant differences seen between groups. For 'time to' first VTE prophylaxis and regional anesthesia placement, prior anticoagulation therapy status was included to factor for significant differences seen between groups; timing of VTE prophylaxis relative to regional anesthesia placement was also included due to the significant intergroup differences in each paired with the inverse relationship identified between these factors.

The results of multivariate regression analysis are shown in Table 3. Lower pain at rest and with activity on day three were significantly correlated with lower pre-block pain ($\beta = 0.31$, $p < 0.001$; $\beta = 0.09$, $p = 0.049$) and older age ($\beta = -0.03$, $p < 0.001$; $\beta = -0.03$, $p < 0.001$).

Table 2
Secondary Outcomes.

Outcome	TEB (n = 188)	ESPB (n = 136)	p-value[A]
Length of Stay, median (IQR)	5.3 (6.5)	7.0 (8.6)	**0.04**
ED, hours	2.2 (2.8)	2.2 (2.2)	0.65
ICU, hours	7.5 (4.8)	7.1 (4.3)	0.54
Hospital, days			
Vasoactive Support, n (%)	45 (23.9)	31 (22.8)	0.81
Block-related	13 (6.9)	0 (0.0)	
Time to First Ambulation, hours median (IQR)	1.7 (4.0)	2.0 (3.8)	0.06
Ventilator Required, n (%)	26 (13.8)	16 (11.8)	0.71
Tracheostomy, n (%)	4 (2.1)	3 (2.2)	1.00
Any Complication, n (%)	18 (9.6)	8 (5.9)	0.32
Hospital/Trauma-Related	0 (0.0)	0 (0.0)	-
ARDS	4 (2.1)	5 (3.7)	0.62
Delirium	1 (0.5)	0 (0.0)	1.00
Encephalopathy	3 (1.6)	1 (0.7)	0.64
Infection	4 (2.1)	3 (2.2)	1.00
Pneumonia	4 (2.1)	0 (0.0)	0.14
VTE	2 (1.1)	1 (0.7)	1.00
Block-related	0 (0.0)	1 (0.7)	0.42
Bleeding or Hematoma	0 (0.0)	0 (0.0)	1.00
Intravascular Injection			
Neuropathy			
Mortality, n (%)	4 (2.1)	3 (2.2)	0.73
Hospital	8 (4.3)	7 (5.1)	0.91
30-day			
Time to First VTE Prophylaxis hours, median (IQR)	21.0 (21.7)	15.4 (16.7)	0.18
Time to Regional Anesthesia hours, median (IQR)	17.2 (18.3)	25.0 (29.1)	**< 0.001**
First VTE Prophylaxis, n (%)	59 (31.4)	91 (66.9)	**< 0.001**
Before Block	128 (68.1)	45 (33.1)	
After Block	1 (0.5)	0 (0.0)	
No VTE prophylaxis			
Time Between VTE Doses hours, median (IQR)	23.9 (12.3)	12.6 (13.7)	**< 0.001**

[A]Bolded p-values represent significant findings at the 0.05 level. Mann Whitney U or Chi Square tests were used to compare medians or frequencies, respectively. Abbreviations: TEB, Thoracic Epidural Block; ESPB, Erector Spinae Plane Block; MME, Morphine Milligram Equivalents; IQR, Interquartile Range; ED, Emergency Department; ICU, Intensive Care Unit; VTE, Venous Thromboembolism

Table 3
Multivariate Regression Analysis.

Outcome	Predictors	Coefficient[B]	p-value[A]	95% CI
NPS Pain, Rest Day 3	Pre-block NPS Pain, Rest	0.31	**< 0.001**	0.21, 0.41
	ESPB Group	−0.42	0.089	−0.90, 0.06
	Age	−0.03	**< 0.001**	−0.04, −0.01
NPS Pain, Activity Day 3	Pre-block NPS Pain, Activity	0.09	**0.049**	0.00, 0.19
	ESPB Group	−0.26	0.368	−0.83, 0.31
	Age	−0.03	**< 0.001**	−0.05, −0.01
		Odds Ratio[C]		
Opioid MME	Pre-block NPS Pain, Rest	1.11	0.075	0.99, 1.24
	ESPB Group	1.13	0.650	0.67, 1.91
	Age	0.93	**< 0.001**	0.91, 0.95
		Hazard Ratio[D]		
Time to First VTE Prophylaxis	Prior anticoagulation therapy	0.64	**0.014**	0.45, 0.91
	Age	1.00	0.491	0.99, 1.00
	ESPB Group	1.04	0.770	0.81, 1.34
	Regional anesthesia beforehand	0.48	**< 0.001**	0.38, 0.62
Time to Regional Anesthesia	Prior anticoagulation therapy	0.70	**0.041**	0.49, 0.99
	Age	1.00	0.559	0.99, 1.00
	ESPB Group	0.92	0.509	0.72, 1.17
	VTE prophylaxis beforehand	0.22	**< 0.001**	0.17, 0.29

[A]Bolded p-values represent significant findings at the 0.05 level, [B]Linear regression, [C]Logistic Regression, [D]Cox Regression. Abbreviations: DF, Degrees of Freedom; SE, Standard Error; CI, Confidence Interval; NPS, Numeric Pain Scale; AIS, Abbreviated Injury Score. ISS, Injury Severity Score; MME, Morphine Milli-Equivalents; VTE, Venous Thromboembolism.

Lower opioid use was more likely among older aged-patients ('0'≤100 MME, '1'>100 MME, OR=0.93, $p < 0.001$). Predictors of prolonged time to first VTE prophylaxis included prior anticoagulation therapy status (HR=0.64, $p = 0.014$) and placement of regional anesthesia beforehand (HR=0.48, $p < 0.001$). Predictors of prolonged time to regional anesthesia placement included prior anticoagulation therapy status (HR=0.70, $p = 0.041$) and first VTE prophylaxis given beforehand (HR=0.22, $p < 0.001$). Multiple pairwise correlation analyses of analgesic and time to outcome variables revealed no co-linear relationship between any predictive variables used in the regression analyses (Supplementary Tables 1 & 2). First VTE prophylaxis before regional anesthesia placement and prior anticoagulation therapy status were both moderately correlated with ESPB grouping ($r = 0.350$ and $r = 0.330$, respectively); all other variables exhibited weak correlations.

Discussion

Thoracic epidural anesthesia has been the preferred mode of analgesia among patients with multiple rib fracture trauma, yet our results suggest that fascial plane blocks offer equally effective analgesia with enhanced safety benefits [14,16–18,20,21]. Our data specifically demonstrate key differences in VTE prophylaxis regime, consistent with increased VTE risk among TEB versus ESPB-treated patients.

During the first five days of local anesthetic infusion, both block groups demonstrated similar, steady improvements in pain without a significant difference in opioid consumption, affirming effectiveness of both techniques. Our results align with existing clinical studies [16, 19–21], which report similar visual analog scale pain and opioid use among groups. Per multivariate regression analysis, higher pre-block pain and younger age, observed in the TEB group, were identified as predictors of increased pain. Consistent with this finding, it is reasonable that PCA opioid infusions were more common in the TEB group. PCA opioid infusions provide complementary analgesia within a multimodal regime and are common in polytraumas wherein patients endure increased pain and injury burden [27]. Thus, higher pain in the TEB group can reasonably be attributed to differences in pre-intervention pain and injury state rather than the anesthetic technique used. Our data indicate provider selection bias towards TEB over ESPB among patients with greater pain, heightened injury, or perceived to be in worse condition. After accounting for this bias, both techniques appear to offer similar analgesia.

Importantly, we found that TEB versus ESPB use significantly affected on VTE prophylaxis regime. For one, the ESPB group received first VTE prophylaxis a median of six hours earlier than the TEB group (16 vs 21 h), not including time spent in the ED. Clinical practice guidelines strongly recommend first VTE prophylaxis within 24 h of injury, given the increased risk of VTE following significant bodily trauma [28]. A recent study has shown that for every hour VTE prophylaxis is delayed following hospital arrival for trauma, the risk of VTE occurrence increases by 1.5% [29]. Additionally, first-line administration of enoxaparin every 12 h is recommended during inpatient care for

traumatic injury to reduce VTE risk [28]. Whereas in our cohort, the median time between VTE prophylactic doses for block placement was approximately 24 h in the TEB group (vs 13 h in ESPB group), almost double the recommended time between prophylactic doses. These results indicate that placement of a TEB over an ESPB can significantly impinge VTE prophylactic regime and may increase VTE risk.

The reason behind these key intergroup differences in VTE prophylaxis is directly related to the anatomical location and associated risks of each block. TEB carries a higher risk of epidural hematoma or hemorrhage complication when anticoagulants are administered concurrently, leading to mandated delays or interruption in VTE prophylaxis during both TEB catheter placement and removal [30]. This is why patients with pre-existing anticoagulants use more commonly received an ESPB over a TEB within our cohort; anticoagulation therapy is also more common among older patients due to increased prevalence of chronic health conditions that enhance VTE risk [31], explaining the greater frequency of older patients (≥65 years) in the ESPB group. Compared to TEB, ESPB, being a fascial plane block, generally has a lower risk of such complications, permitting earlier anticoagulant administration without delay or interruption for block placement. Indeed, four VTE events occurred in the TEB group and none in the ESPB group. Although not statistically significant, likely due to the smaller cohort size, the number of VTE events among the TEB group is clinically relevant, and potentially reflective of regional anesthetic block choice.

Besides reducing risk of thrombotic events, ESPB holds other safety benefits over TEB including enhanced hemodynamic stability. It is well known that TEB can induce blood pressure instability due to vasodilation within the blocked space and reflex vasoconstriction in adjacent unblocked areas [32]. To minimize hypotension with TEB, it is commonplace to administer an opioid adjuvant alongside a reduced dose of local anesthetic. Indeed, about 12.2% of patients in the TEB group received hydromorphone or fentanyl added to the local anesthetic mixture for the epidural infusion. Even with this mitigation strategy, 6.9% of patients in the TEB group required vasoactive support due to block-induced hypotension. Of the 31 patients in the ESPB group who received vasoactive support, only nine received vasoactive support within 24 h of block placement and all these cases were attributed to procedural interventions or pre-existing hypotension. Thus, our findings align with existing trials reporting improved hemodynamic stability among patients who received ESPB versus TEB [16,20,21]. Moreover, it is routine anesthetic practice to place an ESPB under ultrasound guidance to minimize risk of inadvertent injury [33], whereas this technology is not generally used during TEB placement. Using ultrasound guidance significantly mitigates common expected complications such as bleeding/hematoma, nerve injury and LAST [34–36]. Together, these data suggest that an ESPB may be a safer analgesic option for multiple rib fracture trauma than the TEB.

Our results have limitations to consider. Before conducting this analysis, we used very specific inclusion and exclusion criteria to minimize factors confounding for pain resulting in patient groups with similar overall trauma and rib fracture injury severity. This resulted in a smaller cohort size, limiting the generalizability of our findings. Confounding influence is also possible due to baseline group differences in age at admission. However, we performed multivariate regression analyses to elucidate the influence of significantly different potential confounding factors on relevant study outcomes. Given the retrospective design of this study, our analysis relied on data availability and assumption of its accuracy. Lastly, there is time and physician bias to consider. While fascial plan blocks have become increasingly more common in recent years, ESPB placement was not standard practice for rib fracture until the late 2010s, specifically between 2017 and 2019. As such, TEB placement may have been more popular in the early study period, due to enhanced knowledge and experience with this block versus the ESPB. By nature of retrospective design, we did not exclude patients presenting with pre-existing spinal issue or spinal fracture, hemodynamic instability, and pre-existing coagulopathy or anticoagulation use. However, these factors may have influenced physician choice of regional anesthesia technique; indeed, among patients with pre-existing anticoagulation therapy, ESPB was preferred over TEB due to the known association of TEB with increased bleeding/hematoma risk. Finally, we retroactively identified two patients that met all inclusion criteria and each received an ESPB for unilateral rib fracture trauma, but were excluded from the analysis due to death within 48 h and lack of available data to assess the primary outcome, 5-day NPS pain. Neither patient experienced any block complications; the preliminary cause of death was acute hypoxic respiratory failure in both patients, consequent to hypovolemic shock development. Exclusion of these cases in the secondary outcome analysis may have biased our findings.

Despite these limitations, our report results are consistent with those of several prospective RCTs pertaining to the analgesic efficacy of TEB versus ESPB for multiple rib fracture trauma in a broader patient population, including geriatric patients who are often excluded from RCTs, but experience significant burden from rib fracture trauma and risk of VTE. Our analysis also includes clinically relevant insight on the safety of TEB versus ESPB on VTE prophylaxis in a trauma setting.

The implementation of regional anesthetic techniques when appropriate, are strongly advised for patients at high risk for health deterioration following traumatic multiple rib fracture injury [3]. Our findings indicate that the ESPB offers comparable analgesia to the TEB for management of traumatic multi-rib fracture with an enhanced safety profile, both in its application and in its reduced interference with VTE management. Altogether, these preliminary findings support first-line use of ESPB rather than TEB in the setting of unilateral, multiple rib fracture trauma. However, larger-scale prospective research is necessary to further elucidate the effect of TEB use, if any, on risk of VTE incidence in the trauma setting.

CRediT authorship contribution statement

Vincze Sarah: Writing – review & editing, Writing – original draft, Validation, Data curation. **Jyoti Chhabra:** Writing – review & editing, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Conceptualization. **Nishant Merchant:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization. **MacLean Douglas J:** Writing – review & editing, Writing – original draft, Validation, Data curation. **Aseel Walker:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Kevin Finkel:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Data curation, Conceptualization.

Ethics

- This manuscript follows the ethical guidelines stated in Elsevier's Publishing Ethics Policy.
- The work the work described has not been published previously except in the form of a preprint, an abstract, a published lecture, academic thesis or registered report.
- The article's publication is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out.
- All authors have made substantial contributions consistent with authorship denotation.
- AI was not used to generate this work in any capacity.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author contributions

NM, AS, and KF: study conception and design, interpretation of findings, and draft revisions; DJM and SRV: acquisition of data, interpretation of findings, draft development and revisions; JC: statistical analysis and draft revisions. All authors read and approved submission of the final manuscript.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.injury.2026.113435.

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